# BIG 2 EVOLUTIONS & FIRE GROUND SKILLS

## Job Performance Requirements

## NFPA 1010

**6.1.2 6.3.1***

**6.3.2* 6.3.6***

**6.3.9* 6.3.10***

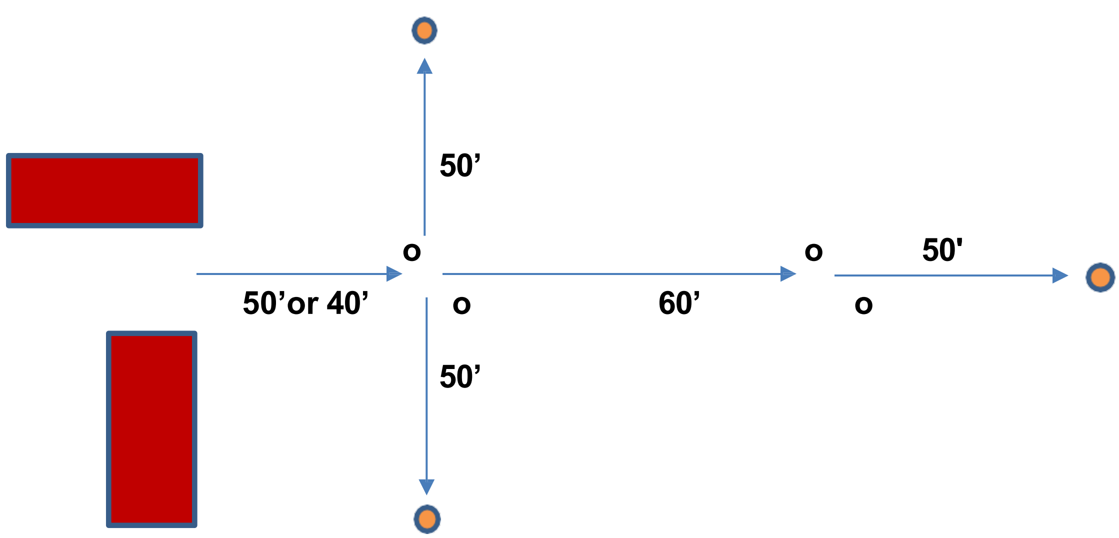
**6.5.1**

# Introduction

Subject: Big 2 & Fire Ground Skills

Note: Setup

1. **PPE, SCBA, HOSE shall be the 1st part.**
2. The layout shall be as follows:
   a. The apparatus will be parked at a designated point.
   b. **SCBA** shall be seat mounted or compartment mounted **(May NOT be staged on the ground).**
   c. The attack hose lines shall be loaded with 150 feet of 1¾ inch hose, using the Flat, Minuteman, and Triple Layer loads. The hose shall be properly loaded with a nozzle attached. The hose shall NOT be hanging out of the hose bed in a manner which would allow the hose to fall from the apparatus while traveling to an emergency.
   d. The 1st entry point, of a structure, will be no less than forty (40) feet from the apparatus (Rear or Side for cross lays). The entry will be thirty-six (36) inches wide.
   e. Two (2) cones will be positioned fifty (50) feet from the edge of the entry, one (1) cone on each side.
   f. The 2nd entry point, inside the structure, will be sixty (60) feet from the 1st entry point. The entry will be thirty-six (36) inches wide.
   g. The 3rd and final cone will be positioned fifty (50) feet from the 2nd entry point.
3. Diagram:

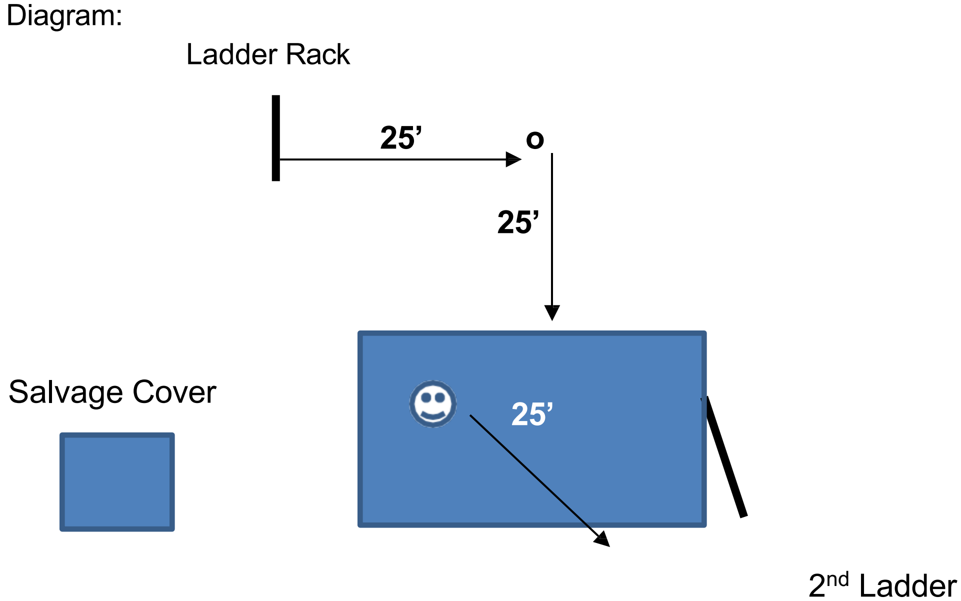


**Note:** If engine is forty (40) feet from entry, hose load may be fully extended to either side of entry and that will not be considered entering the IDLH environment. Exception, if hose is fully extended through the entry on initial pull that will be considered entering the IDLH environment.

4. **Ladder, Search & Rescue shall be the 2nd part.**
5. The layout shall be as follows:
    a. The ladder rack will be positioned at a designated point.
    b. A twenty-four (24) foot extension ladder, and a fourteen (14) foot roof ladder will be secured with locking devices on the ladder rack.
    c. A cone will be positioned perpendicular to the ladder rack and building, and twenty-five (25) feet from the ladder rack and building.
    d. A 2nd ladder will be positioned and secured at a designated point on the building. This ladder will be utilized for climbing to the 2nd floor entry point/window, to complete the Search & Rescue portion of the evolution.
    e. A rescue victim will be positioned at a designated point on the ground floor inside the building, and twenty-five (25) feet from the exit point/door. If the building configuration does not allow 25 feet between the victim and exit, place tape or another marker beyond the exit creating the 25 feet to the threshold.

<u>Note:</u> Candidate must walk a minimum of fifty (50) feet from the ladder extended to the 2nd ladder. The 2nd ladder does NOT need to be positioned fifty (50) feet from the 1st ladder, in order to meet this requirement. A cone can be placed twenty-five (25) feet away (in the opposite direction) from the 1st ladder. The candidate will walk to and around the cone, then proceed to the 2nd ladder totaling fifty (50) feet.

6. Diagram:



7. **The Fire Ground Skills shall be the 3rd part.**
8. All Equipment shall be stored on salvage cover or on fire Apparatus. If the candidate must travel more than fifty (50) feet, they will be allowed to move equipment to a staging area close to the assigned task station. It is acceptable to have a second salvage cover with all the tools, equipment, hose, and appliances at a second location, if the training grounds are large.
9. The Fire Ground Skills consists of four (4) components: ERG, Knot, Two (2) Practical Skills. Each component is worth twenty-five (25) points.
10. The ERG (Emergency Response Guide) exercise is an open book, 5 question, multiple choice, written test. This Performance Objective/Skill shall be completed within 20:00 minutes. The Examiner will decide if this test is conducted prior to the outside practical evaluations, or after completing the outside skills.
11. The Knot skill shall be performed between the Big 2 evolutions. All knots shall be tied within 2:00 minutes, regardless if tying a hand knot or a tool for hoisting. A rope shall be affixed to an elevated location at one end, and the other end shall be utilized to tie tools for hoisting and hand knots. A second rope shall be affixed to an elevated location at one end with the other end left in a rope bag and not accessible. **This rope shall be utilized for tying tools in-line (in the middle of the rope).** A third rope shall be located on the ground for tying hand knots and tag lines. It is recommended the diameter should be a different size for tying the Becket Bend as to not confuse the student during testing. All knots shall be dressed upon completion (Loose knots are NOT functional). Safeties shall be tied and snug against the knot.

    Note: A valuable resource is "animatedknots.com." (Although the BFST authorizes the use of additional resources, the BFST does not endorse this website.)

12. The remaining two (2) components shall involve any combination of the following performance objectives:
    a. Forcible Entry
    b. Ground Ladders
    c. Hose, Tools, and Appliances
    d. Salvage and Overhaul
    e. Ventilation
    f. Fire Extinguishers
    g. Exiting a Constricted Opening
13. Individual skills shall be completed within 4 minutes and 40 seconds (4:40).
14. For IDLH conditions, candidates shall go on air to perform the skill.

# Performance Objective

Subject: Big 2

Task: PPE, SCBA, HOSE

Note: Running or walking backwards is prohibited throughout the entire evolution. When operating a charged hose line, open/close the bale of the nozzle slowly/fully. Any time a candidate enters or exits a structure, a radio transmission (PAR) shall be completed.

Procedure:

1. Candidate will perform pre-evolution functionality and safety checks, on PPE and SCBA. When completed, place the SCBA in the assigned jump seat and stage your PPE for testing.
2. Testing begins when the evaluator assigns a pre-connected hose load for the candidate to utilize during the evolution. The candidate will inspect and verify the hose load, then the evaluator will assign a cone knockdown sequence.
3. The timed portion of the test shall be completed within 4 minutes and 40 seconds (4:40). Time begins when the candidate touches and dons their PPE. All attachments (Buttons, Clips, Snaps, Velcro, Zippers) shall be fastened.
4. Board the apparatus, don and turn on the SCBA verbalizing: bottle pressure, low pressure alarm, remote pressure, PASS activated. You will don the face mask and execute a proper seal check (You can don the face mask and turn on the SCBA at the 1$^{st}$ entry point).
5. After donning and securing PPE properly, dismount the apparatus and proceed to the assigned pre-connect hose load.
6. Pull the entire hose load completely out of the hose bed. Deploy the hose as necessary, to the 1$^{st}$ entry point. Lay the hose in a manner to avoid excessive kinks when the hose is charged with water.
7. Before breaking the plane of entry: Signal for water and bleed the charged hose line; Make radio transmission to enter (PAR); Go on air.

Note: When opening/closing the bale of the nozzle, it shall be done slowly/fully to prevent water hammer and possible damage to the apparatus. The BFST recommends utilizing a three (3) second count to ensure this is accomplished.

8. Enter the structure and knock down the cones in the order assigned. Do NOT advance the hose line while water stream is flowing.

   Note: If it is determined the water stream is inadequate due to excessive kinks in the hose and the decision is made to leave the structure, a radio transmission to exit (PAR) shall be made. After correcting the issues, a radio transmission to enter (PAR) shall be made. It is imperative to transmit PAR whenever entering and exiting a structure. This can be accomplished while walking and it is NOT necessary to stop moving while transmitting.

9. After knocking down both cones, close the bale, advance the hose line to the 2nd entry point (inside the structure), and knock down the final cone. Time stops after final cone falls.
10. You are still graded on the following: make radio transmission (PAR) confirming fire is extinguished, PPE, doffing mask.

# Performance Objective

Subject: Big 2

Task: Ladder, Search & Rescue

Note: Proper body mechanics shall be used when lifting and lowering ladders. Three (3) ladder guards will be utilized for this evolution. Candidates shall always remain in contact with the ladder while: Deploying; Raising; Extending the fly section; Until properly relieved and heeled by ladder guards. Any time a candidate enters or exits a structure, a radio transmission (PAR) shall be completed.

Procedure:

1. Candidate will perform pre-evolution functionality and safety checks on both ladders.
2. The timed portion of the test shall be completed within 4 minutes and 40 seconds (4:40). Time begins when the candidate touches any part of the ladder or locking device.
3. Disengage the locking devices securing the ladder on the apparatus.
4. Remove the roof ladder and place it on the ground, ensuring it is not directly exposed to the exhaust of the apparatus.
5. Remove the extension ladder, utilizing a low shoulder carry and with the bed section against your body. Face the butt end of the ladder and carry with the butt slightly lowered.
6. Verify and verbalize for any overhead or ground obstructions in the area.
7. Carry the ladder twenty-five (25) feet toward the cone, walk around the cone and advance twenty-five (25) feet toward the building at the designated location (a left and right turn shall be performed).
8. Stop just short of the building, ensuring NOT to hit the building with the butt of the ladder.

9. Kneel and remove the ladder from your shoulder, place the ladder on its beam and lay the ladder flat, with the bed section down on the ground.
10. Position yourself at the tip of the ladder, slide the butt of the ladder into the building, and raise the ladder to a vertical position with the fly section against the building.

    Note: When carrying the ladder, you may “stick/spike” the butt into the building and directly raise the ladder while standing. Replaces steps 9 and10.

11. Reposition the butt a safe distance from the building.
12. Grasp the halyard and extend the fly section completely, without touching the building. Ensure both dogs are locked and press the tips into the building.
13. Grasp the excess halyard lying on the ground, lift the ladder leaving the tips on the building for stability, and reposition the butt approximately six (6) feet from the building. This shall be done without walking backwards.
14. Place a foot on the bottom rung or against the beam, then pass the excess halyard through the ladder (approximately waist high) between the rungs.
15. Grasp the excess halyard underneath the rung it was passed through and form a bight. Pull the slack out of the halyard and keep it taut.
16. Tie a Clove-Hitch on a bight around the upper rung the halyard was passed through, with a Safety.
17. Rotate the ladder so the fly section is positioned out.
18. After rotating the ladder, a proper climbing angle will be achieved and verified by utilizing the following methods:
    a. The toes of both feet should touch the butt;
    b. Stand erect with arms extending straight out and at shoulder level;
    c. The hands should be positioned between the beams as if grasping a rung;
    d. Maintain this position without leaning to compensate an incorrect angle.
19. Verify and verbalize the following:
    **a. Four points of contact;**
    **b. Both dogs are locked;**
    **c. Halyard is tied and secured;**
    **d. Proper climbing angle;**
    **e. Ladder is safe to climb.**
20. After verifying the ladder, heel the ladder by positioning yourself using one of the following methods:
    a. **Underneath:** Stand with feet offset for stability, grasp both beams at eye level below a rung, pull and apply constant pressure against the building while looking straight ahead.
    b. **Outside:** Stand with one foot placed on the bottom rung, grasp both beams and press the ladder against the building.

21. Instruct ladder guards to take control of the ladder (Do NOT remove your hands until guards grasp the ladder).
22. Walk fifty (50) feet, proceed to the 2nd ladder, and instruct ladder guard to heel the ladder. Verify ladder and only verbalize **“ladder is safe to climb.”**
23. Verbalize remote pressure, don your face mask and execute a proper seal check. Before breaking the plane of entry, make radio transmission (PAR) and go on air.
24. Ascend the ladder with a tool to the 2nd floor. Slide the free hand on the backside of the beam, maintaining continuous contact with the beam. Before dismounting the ladder onto the windowsill, sound the floor and enter the building.
25. Conduct a search inside the building and descend to the ground floor.
26. When the victim is located, make radio transmission, and notify command of rescue attempt.
27. Remove the victim from the building, using one of the following approved rescue methods:
    a. **1st Method:** Stand behind the victim, place your arms under the victim’s arms, then drag the victim out of the building.
    b. **2nd Method:** Stand behind the victim, place a rope or webbing around the victim’s chest and under the arms, then drag the victim out of the building.
28. Exit the building, ensuring the victim is completely removed from the structure. Time stops when victim’s feet crosses the threshold.
29. You are still graded on the following: gently lay the victim down so the head does NOT hit the ground, make radio transmission (PAR) confirming victim rescue, PPE, doffing mask.

**NOTE: Using any portion of the Victims Head or Neck to lift them for removal from the building, or to lower them, once exited from the building is not acceptable and will result in a CRITACLE FAILURE !!!!!!**